RADIATION DOSE OPTIMISATION · CORONARY CT ANGIOGRAPHY

CT Dose AI

Predicting $\mathrm{CTDI_{vol}}$ and DLP — and learning the lowest dose that still gives a diagnostic image.

0.9%
$\mathrm{CTDI_{vol}}$ error, cross-validated

8.2%
DLP error, cross-validated

≈40%
dose saving at 100 kV at equal iodine CNR

2
learning loops: phantom + clinical

Thodoris Fragos · 23 September 2026
50 coronary CT angiography exams · Python / scikit-learn / Streamlit
github.com/Thodoris1657/CT-Dose-AI-Algorithm

What we were asked to build

> “We want the AI programme to take **BMI, kVp, mAs and scan length** and predict **CTDI and DLP**. Then to see whether it can suggest **what I should change** in kVp, mAs or scan length **to reduce CTDI and DLP**. We consider BMI fixed — it depends on the patient’s height and weight.”

Problem 1 — prediction

What dose will this protocol deliver? A supervised-learning problem with a strong physical structure, and only 50 exams to learn from.

Problem 2 — decision

What should I change? Minimising dose alone is trivial and dangerous — scan with almost nothing. The real question is **the lowest dose that is still diagnostic**, and image quality is not in the data at all.

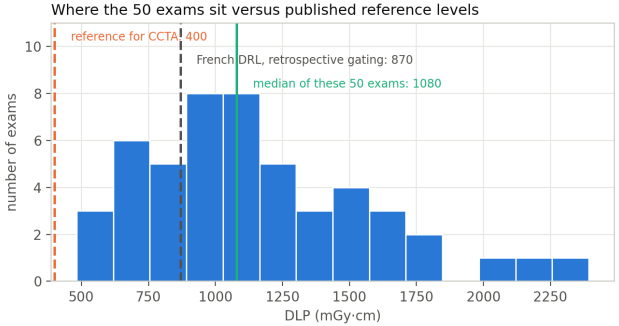
Why it matters: the current dose level

CCTA is one of the **highest-dose routine CT exams**, often in relatively young patients.

Published reference levels: **170–400** mGy·cm for prospective gating, **870–1060** for retrospective.

These 50 exams: median **1080** mGy·cm, maximum **2393**.

Every exam was done at **120 or 140 kV**, at every BMI from 18 to 59.



The dose level is consistent with retrospective gating at a fixed tube voltage. That is exactly where the headroom is.

THE DATA

50 exams, 7 columns — and four findings from the audit

Field	Median	Range
BMI (fixed)	27.2	18.3–58.8
kVp	120	120 (46) · 140 (4)
mAs	756	375–1345
Scan size (cm)	29.8	22.4–44.2
$\mathrm{CTDI}_{\mathrm{vol}}$ (mGy)	47.5	21.1–105
DLP (mGy·cm)	1080	482–2393

Source: a spreadsheet printed to PDF. The app parses it directly, so re-imports need no manual typing.

$\mathrm{CTDI}_{\mathrm{vol}}$ is deterministic: 0.0605 mGy per mAs at 120 kV, 0.0894 at 140 kV — a scanner constant, not a correlation.

"SCAN SIZE CM" is not the scan length: it tracks BMI ($r = 0.89$) but not DLP/CTDI ($r = -0.23$). The implied irradiated length is 23.2 ± 2.2 cm.

Five exams give 7% less dose per mAs — a different collimation, pitch or scan mode that is not recorded anywhere.

Two pairs of duplicate rows — kept in the same cross-validation fold so they cannot inflate the score.

The core insight: the dose is physics, not a pattern

$$\mathrm{CTDI_{vol}} = k \cdot \mathrm{mAs} \cdot (\mathrm{kVp}/120)^{n}$$

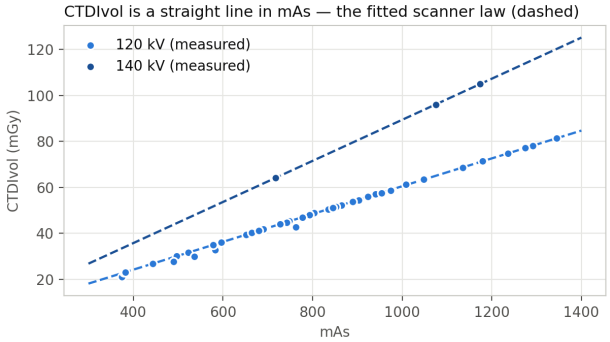
$$\mathrm{DLP} = \mathrm{CTDI_{vol}} \times \mathrm{length}$$

Fitted to this scanner: $k = 0.0605$ mGy/mAs, $n = 2.53$ (physics expects 2.5–3.0).

The **mAs exponent is fixed at 1**, not fitted — dose is linear in tube charge. That is what lets the model extrapolate safely.

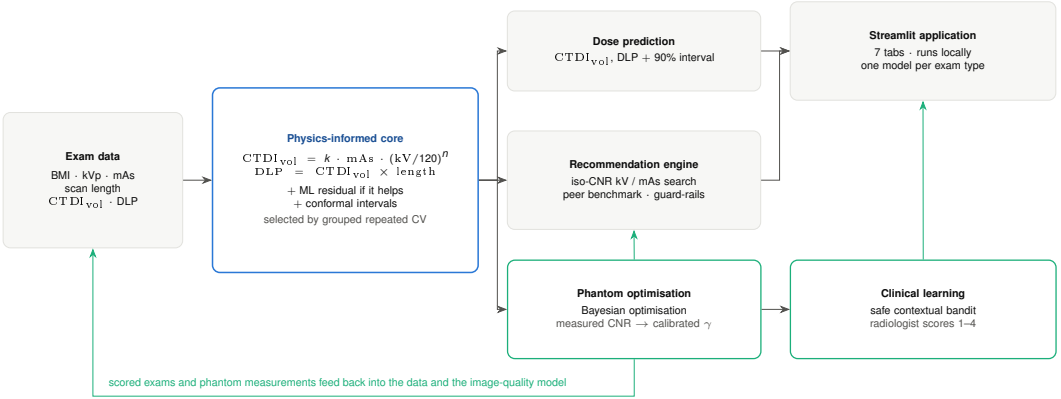
Robust (Huber) fitting, so the five deviant exams do not distort the constant.

Everything is fitted in **log space**: the drivers multiply, they do not add.



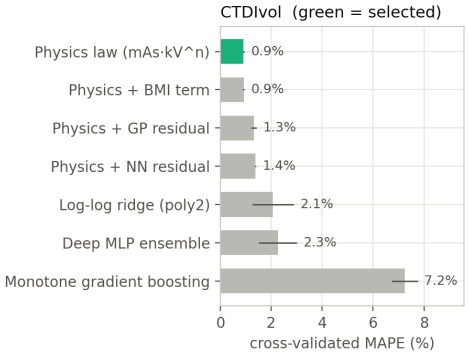
Dashed lines are the fitted law, not a trend line through a cloud.

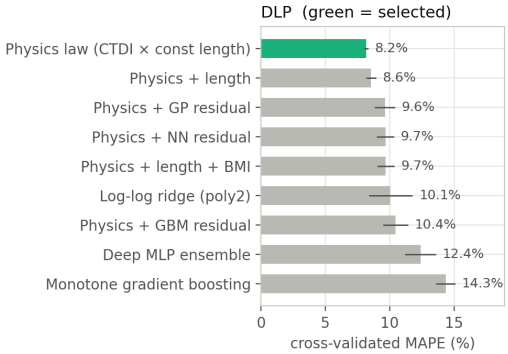
One physics core, three layers



Blue = built from the supplied data. Green = the two loops that go and acquire the image-quality information the data does not contain.

Nine models competed. Physics won — deep learning did not.





Protocol: repeated grouped 5-fold CV, 10 repeats; duplicates locked into the same fold.

Selection: 1-standard-error rule — the simplest model within one SE of the best.

Deep MLP ensemble: 2.3% and 12.4% — 2.5× and 1.5× worse than the physics law.

With 48 unique rows and a known functional form, that is the expected result, not a surprise.

The simple model is the answer — the ML is everywhere else

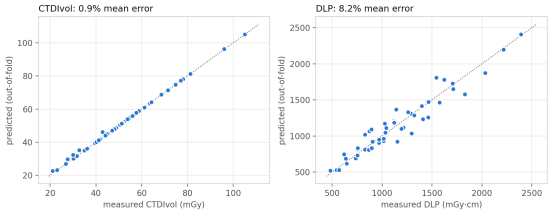
What machine learning contributes here

An honest referee: grouped repeated cross-validation and the 1-SE rule, so no model wins on a fluke.

Calibrated uncertainty: conformal intervals — CTDI $\times[0.93, 1.00]$, DLP $\times[0.82, 1.25]$ at 90%, measured coverage 94%.

Knowing its limits: every input outside the training range is flagged, and 80/100 kV is marked as physics extrapolation.

Constrained optimisation and **two active-learning loops** — the parts that actually reduce dose.



Every point predicted by a model that never saw it. The CTDI outliers are the five exams with the unrecorded scan mode; the DLP spread is the unrecorded scan length.

“What should I change?” — minimum dose at constant image quality

$$\text{noise} \propto 1/\sqrt{\mathrm{CTDI_{vol}}}$$

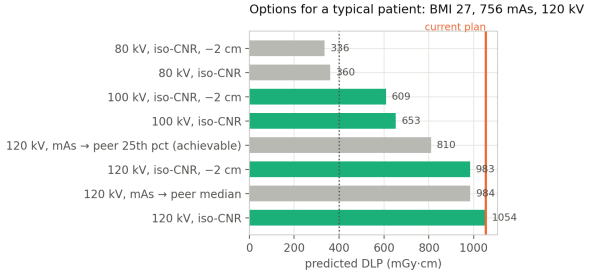
$$\text{iodine contrast} \propto \mathrm{kVp}^{-\gamma},\ \gamma \approx 1.35$$

$$D_{\mathrm{new}} \geq D_{\mathrm{old}}\,\tau^2\,(\mathrm{kV_{new}}/\mathrm{kV_{old}})^{2\gamma}$$

Lower kV is the dominant lever in contrast-enhanced CT: at 100 kV the same iodine CNR needs ≈40% less dose.

Guard-rails

- Minimum kV by BMI (80 / 100 / 120)
- Tube output limit on mAs
- Peer benchmark from your own data
- Length savings reported separately



THE SOLUTION · WORKED EXAMPLE

A typical patient: BMI 27, 120 kV, 756 mAs

−39%

$\mathrm{CTDI_{vol}}$: 45.7 → 28.0 mGy

−42%

DLP: 1056 → 609 mGy·cm (with 2 cm less coverage)

100%

of the current estimated iodine CNR retained

100 kV

735 mAs — the proposed protocol

How it was derived

The mAs is raised slightly to compensate the kV drop exactly at constant iodine CNR; the dose still falls because tube output scales as $\mathrm{kV}^{2.53}$. Scan length acts directly on DLP.

What the tool says about itself

“100 kV does not appear in the training data — this estimate relies on the fitted physics kV-law ($n = 2.53$). Validate with a phantom or the first scans.” The tool declares its extrapolations.

THE GAP

But how low can we actually go?

The data contains no information about image quality. Physics can say what keeps CNR constant; it cannot say how much CNR is enough for a confident read. So the system goes and measures it — in two places.

1 · On the phantom

Bayesian optimisation. The app proposes the next scan, you measure CNR, it proposes the next one. Roughly 8–20 scans instead of a full grid.

Output: the lowest-dose setting that meets your CNR target, plus a γ calibrated on your own scanner.

2 · In clinical routine

Safe contextual bandit. One proposal per patient, a radiologist approves or overrides it, then scores the images 1–4.

Output: the lowest dose that stays diagnostic for each BMI — learned from your own readers, one small step at a time.

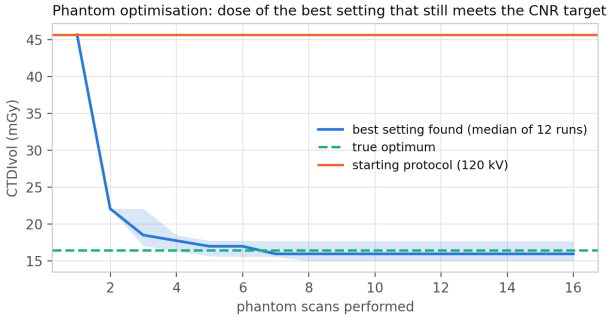
Bayesian optimisation: minimise dose subject to measured CNR

Surrogate:
$\log \mathrm{CNR} = b_0 - \gamma \log(\mathrm{kV}/120) + \delta \log \mathrm{CTDI}_{\mathrm{vol}}$
+ Gaussian-process residual, with physics priors ($\gamma \approx 1.35$, $\delta \approx 0.5$). The priors make it useful after 2–3 scans instead of 30.

Acquisition: expected dose saving × probability of meeting the CNR target.

Dose is known analytically from the fitted scanner law, so every scan is spent on the unknown — image quality.

Result in simulation: the true constrained optimum after ≈7–8 scans, within ∼8% of it, target met in 10 of 10 runs.



12 simulated campaigns on a synthetic phantom; band = 10th–90th percentile.

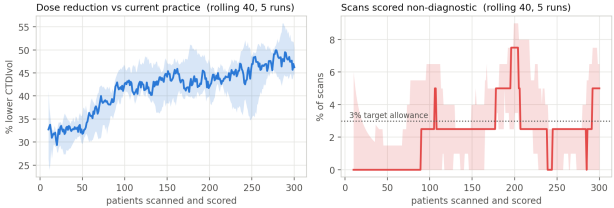
A bandit that learns from radiologist scores — under supervision

Context BMI · **actions** kV × dose level · **reward** diagnostic score 1–4 ($\geq$ 3 acceptable).

Model: Bayesian logistic regression of P(acceptable) on BMI, log CTDI and log kV, with physics-informed priors. One shared model, so every exam teaches every protocol.

Decision: Thompson sampling — the lowest-dose allowed action whose sampled P clears 0.97. Exploration shrinks by itself as evidence builds.

In simulation: −32% dose immediately, −45 to −50% later; 3–5% non-diagnostic, matching the 0.97 target the department sets.



5 runs × 300 simulated patients with a synthetic reader; median and 10th–90th percentile.

SAFETY

Four rails, always on

1 · Minimum kV by BMI

80 kV below BMI 25, 100 kV for 25–30, 120 kV above 30. No proposal can go below it — photon starvation is never traded for dose.

2 · Pessimistic probability floor

The 10th-percentile posterior P(acceptable) of a proposal must exceed 0.85. Uncertainty counts against a protocol, never for it.

3 · Small, evidenced steps

A proposal may be at most 15% below the lowest dose already proven acceptable at a similar BMI. The dose walks down; it never jumps.

4 · A human decides

A radiologist approves or overrides every proposal, and every override is logged. The algorithm proposes; it never acts on its own.

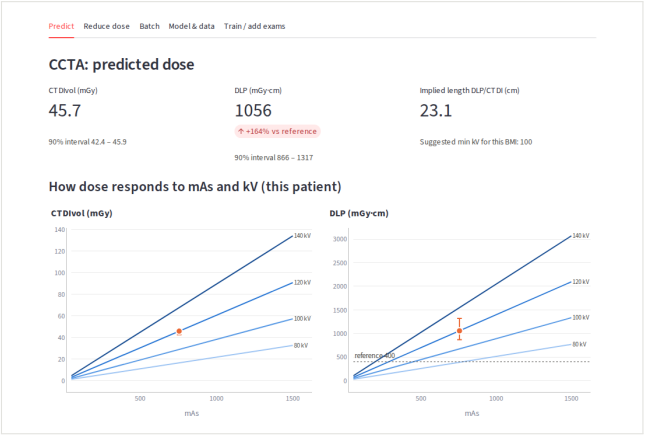
Plus: out-of-domain warnings on every prediction, 13 automated tests including “an obese patient is never offered a sub-120 kV protocol”, and all data stays on the local machine.

A local application — seven tabs, one command to start

Tab	Purpose
Predict	dose + 90% interval, reference comparison
Reduce dose	ranked options at constant CNR
Phantom (BO)	next scan to perform, calibrated γ
Clinical learning	propose → approve → score → learn
Batch	whole file at once, with error report
Model & data	model card, CV table, audit
Train / add exams	any other examination, its own model

Runs offline on the department PC. Nothing is uploaded.

```
python app.py
```



The Predict tab: dose with its 90% interval, the reference-level comparison, and how dose responds to mAs at each kV for this patient.

THE SYSTEM

Every scan makes the next one better

1 · Phantom → engine

The campaign measures γ on your scanner, and it replaces the literature value in the recommendation engine and in the bandit's action set.

2 · Readers → engine

Each scored exam updates the acceptability model. "Reduce dose" then shows the lowest dose *proven* acceptable at that BMI — local evidence, not assumption.

3 · Console → dataset

Scored exams with console doses are appended to the training data. The next training run re-runs the full model competition on more data.

The dataset that started as 50 rows in a PDF grows by itself through routine use — and if a flexible model ever starts winning the competition, the pipeline will select it automatically.

RESULTS

Where we stand

Question	Answer	Evidence
Predict $\mathrm{CTDI_{vol}}$?	Yes — 0.9% error, R^2 0.998	repeated grouped CV, 50 exams
Predict DLP?	Yes — 8.2% error, R^2 0.92	limited by the unrecorded scan length
Does deep learning help?	No — 2.3% and 12.4%	identical protocol for all nine models
Say what to change?	Yes — ranked options at constant CNR	iso-CNR physics + guard-rails
How much dose could go?	≈40% at 100 kV, ≈65% at 80 kV (slim)	physics estimate, phantom to confirm
Find the optimum experimentally?	Yes — ≈8 phantom scans	12 simulated campaigns
Keep learning safely in routine?	Yes, with approval at every step	1500 simulated patients

What the simulations prove: that the algorithms converge, respect their constraints and behave safely. **What they do not prove:** clinical benefit. The real numbers depend on your scanner, your reconstruction and your readers — which is what the phantom campaign and the pilot are for.

Five questions for the data provider

What exactly is “SCAN SIZE CM”? It tracks BMI, not the irradiated length. The true z-coverage should take DLP error from $\sim$8% towards $\sim$1%.

What differs in rows 2, 9, 22, 41, 45? They deliver 7% less dose per mAs — probably collimation, pitch or scan mode.

Are rows 28/43 and 29/42 really two patients each? They are identical in every field.

Is the protocol retrospectively gated? If prospective triggering is feasible at the patient’s heart rate, it saves more than any kV or mAs change — but it is outside the four parameters we were given.

Which fields can be exported routinely? Scan length · gating mode · collimation and pitch · heart rate · reconstruction level · a 1–4 diagnostic-quality score.

NEXT STEPS

From 50 rows in a PDF to a department-wide tool

Stage	Action	Outcome
1 · Data	Export the extra fields for existing exams and the next 100–200 cases	DLP error towards 1–2%
2 · Phantom	One campaign with the physicist: 10–20 scans, entering measured CNR	Site-specific γ, evidence-based low-kV protocol
3 · Pilot	20–30 patients at the phantom-derived protocol, each scored 1–4	First real acceptability data; go / no-go
4 · Routine	Clinical-learning loop on conservative settings	Gradual, evidenced dose reduction
5 · Expansion	Brain, chest, abdomen — each gets its own model	Department dose-optimisation platform

Thank you

Questions, and the phantom slot, are the next step.
github.com/Thodoris1657/CT-Dose-AI-Algorithm